Silpa Vadakkeeveetil Sreelatha

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Research Interests

Responsible AI • Generative Models • Mechanistic Interpretability • Fairness & Bias Mitigation • Text-to-Image Generation • Computer Vision

Technical Skills

Programming: Python
Frameworks & Libraries: PyTorch, Diffusers, Hugging Face Transformers, Stable Diffusion (SD/SDXL/FLUX), NumPy, SciPy, scikit-learn
Tools: Weights & Biases, Git, SLURM (HPC / Isambard-AI), LaTeX
Infrastructure: Large-scale distributed training on HPC clusters (120,000 GPU hours, Isambard-AI AIRR)
Platforms: Linux/Ubuntu

Selected Projects

RAIGen: Rare Attribute Identification in Text-to-Image Models | *ICML 2026* — 2025 – 2026

- Built a sparse autoencoder (SAE) probing framework on diffusion model feature spaces to discover minority and rare attribute representations, surfacing concepts that standard generation systematically under-represents.
- Enables targeted, retraining-free fairness interventions; validated on FLUX and Stable Diffusion XL across multiple demographic attributes.

RespoDiff: Responsible Text-to-Image Generation | *NeurIPS 2025* | Code — 2024 – 2025

- Designed a dual-module bottleneck transformation that locates and neutralizes bias-encoding concepts inside diffusion model internals.
- Improves responsible and semantically coherent generation by **∼20%** across diverse, unseen prompts; outperforms prior debiasing baselines on demographic fairness metrics (Winobias, CLIP alignment) without degrading image fidelity.

DeNetDM: Debiasing by Network Depth Modulation | *NeurIPS 2024* | Code — 2023 – 2024

- Proposed a depth-based modulation strategy to disentangle spurious correlations from core semantics in image classifiers; outperforms existing debiasing techniques by **5%** on both synthetic and real-world datasets without requiring bias annotations.

Research Experience

PhD Researcher, ELLIS PhD Student — June 2023 – Present

University of Surrey, UK (visiting: Pioneer Centre for AI, Copenhagen, Aug 2025 – Feb 2026)

- Developed SAE-based probing methods to uncover minority and rare attribute representations in diffusion model latent spaces, enabling retraining-free fairness interventions (*RAIGen*, ICML 2026).
- Designed a dual-module bottleneck transformation that identifies and nullifies bias-encoding concepts in diffusion internals at inference time, achieving state-of-the-art demographic fairness (*RespoDiff*, NeurIPS 2025).
- Awarded 120,000 GPU hours on UK national HPC (Isambard-AI AIRR) to scale T2I trustworthiness experiments.

Independent Researcher — April 2022 – May 2023

- Proposed network depth modulation to disentangle shortcut correlations from core semantics in image classifiers, improving worst-group accuracy across distribution shifts without requiring bias annotations (*DeNetDM*, NeurIPS 2024).

Researcher, Deep Learning & AI — July 2019 – April 2022

Tata Research Development and Design Centre (TCS Research), Pune, India

- Built cross-modal retrieval systems (Siamese BERT, LXMERT, ViLBERT) achieving state-of-the-art on the CVPR 2018 Advertisement Understanding benchmark (ACL 2020).
- Developed a data-free knowledge distillation pipeline that transfers knowledge from a large teacher model to a compact student without access to original training data.
- Co-authored US Patent US11699275B2 on visio-linguistic understanding using contextual language model reasoners.

Selected Publications & Patents

*Equal contribution.

- **RAIGen: Rare Attribute Identification in Text-to-Image Generative Models. Silpa Vadakkeeveetil Sreelatha**, D. Wang, S. Belongie, M. Awais, A. Dutta. **ICML 2026.**
- **RespoDiff: Dual-Module Bottleneck Transformation for Responsible & Faithful Text-to-Image Generation. Silpa Vadakkeeveetil Sreelatha**, S. Nag, M. Awais, S. Belongie, A. Dutta. **NeurIPS 2025.**
- **DeNetDM: Debiasing by Network Depth Modulation. Silpa Vadakkeeveetil Sreelatha***, A. Kappiyath*, A. Chaudhuri, A. Dutta. **NeurIPS 2024.**
- **Self-Supervised Enhancement of Latent Discovery in GANs.** A. Kappiyath*, **Silpa Vadakkeeveetil Sreelatha***, S. Sumitra. **AAAI 2022.**
- **Understanding Advertisements with BERT.** K. Kalra, B. Kurma, **Silpa Vadakkeeveetil Sreelatha**, M. Patwardhan, S. Karande. **ACL 2020.**
- **Method and System for Visio-Linguistic Understanding using Contextual Language Model Reasoners.** Kurma S. S. B., Kalra K., **Silpa Vadakkeeveetil Sreelatha**, Patwardhan M., Karande S. **US Patent US11699275B2, 2023.**

Invited Talks

- "Towards Responsible Text-to-Image Diffusion Models," DTU Compute, Technical University of Denmark, Copenhagen, Denmark, January 2025.

Education

ELLIS Ph.D. in People-Centred Artificial Intelligence June 2023 – July 2027
University of Surrey, United Kingdom Advisors: Anjan Dutta, Serge Belongie, Muhammad Awais

Visiting Ph.D. Student — ELLIS Research Exchange Aug 2025 – Feb 2026
Pioneer Centre for Artificial Intelligence, University of Copenhagen, Denmark Supervisor: Prof. Serge Belongie

M.Tech in Machine Learning and Computing — CGPA: 9.31/10.00 July 2017 – June 2019
Indian Institute of Space Science and Technology (IIST), Kerala, India Supervisor: Sumitra S

B.Tech in Computer Science and Engineering — CGPA: 7.85/10.00 Aug 2012 – May 2016
Government Engineering College, Sreekrishnapuram, Kerala, India

Honors & Awards

2026 – Awarded 120,000 GPU hours on Isambard-AI AIRR (project: TrustDiff).
2025 – ELSA Mobility Grant, ELLIS Symposium, Warsaw, Poland.
2023 – Selected for the ELLIS PhD Program (≈5% acceptance rate).
2018 – Finalist, Microsoft AI Challenge.
2017 – Department of Space Fellowship for M.Tech at IIST.
2017 – Ranked 1,349 / 96,878 (98.6th percentile) in All-India GATE (CS&E).

Professional Service & Teaching

Conference Reviewer: ICLR (2024–2026), NeurIPS (2023–2026), CVPR (2024–2025), ICML (2024), ICCV (2025).
Workshop Reviewer: WiCV at CVPR 2023; SafeGenAI at NeurIPS 2024.
Teaching: Co-supervisor, MSc dissertations on controllable image editing (2024–2025) and on building robust classifiers using generative models (2022–2023); Teaching Assistant, Applied Machine Learning (2023, 2024), University of Surrey; mentored ∼50 M.Tech students on course projects in diffusion models.